Clinic Appointment and Patient Management System

A Final Group Project

Presented to the Faculty of the

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Project Scenario

A private medical clinic wants a database system to manage patients, doctors, appointments, consultations, and prescriptions. The clinic has several doctors with different medical specialties. Each doctor has a unique doctor’s ID, full name, specialization, license number, email address, contact number, and consultation fee. A doctor belongs to one specialty, while a specialty may have several doctors. Patients who visit the clinic must be registered in the system. Each patient has a unique patient ID, full name, date of birth, gender, address, contact number, email address, and emergency contact information. Patients schedule appointments with doctors. An appointment contains the appointment ID, appointment date, appointment time, reason for visit, appointment status, and notes. A patient may have many appointments over time, while a doctor may see many patients. However, a doctor cannot have two appointments scheduled for the same time. When a patient attends an appointment, the doctor conducts a consultation. The consultation records the patient's symptoms, diagnosis, medical notes, treatment recommendations, and consultation date. Not every scheduled appointment necessarily results in a consultation because an appointment may be cancelled or the patient may fail to attend. Doctors may prescribe medications to patients as part of a consultation. A prescription records the medication, dosage, frequency, duration, and instructions. A consultation may result in multiple prescribed medications. The clinic wants the system to provide information about a patient's appointment history, diagnoses, prescriptions, and upcoming appointments. It should also allow staff to identify a doctor's daily schedule.

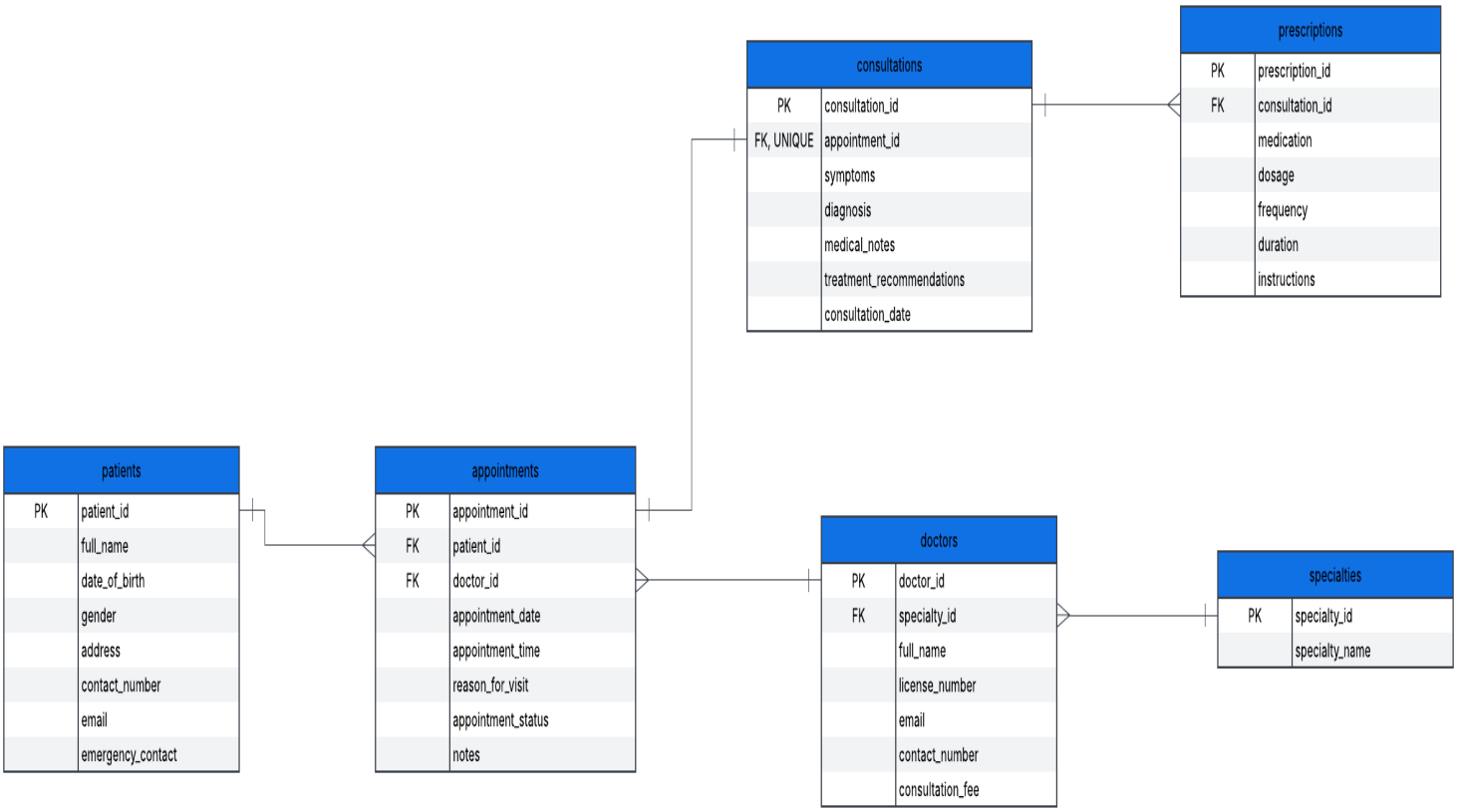
Business Rules and Requirements

- Every patient must have a unique patient ID.
- Every doctor must have a unique doctor’s ID.
- A doctor belongs to one specialty.
- A specialty may have multiple doctors.
- A patient may have many appointments.
- A doctor may have many appointments.
- An appointment belongs to one patient and one doctor.
- A doctor cannot have overlapping appointments.
- An appointment may be Scheduled, Completed, Cancelled, or No-show.
- A completed appointment may result in a consultation record.
- A consultation is associated with one appointment.
- A consultation may include multiple prescribed medications.
- The system should maintain the patient's medical and appointment history.

Required Output

1. Display All Patients and Their Appointment History
2. Display Doctors' Daily Schedule
3. Display Patient Consultations, Diagnoses, and Prescriptions

Entity Relationship Diagram



The Entity-Relationship Diagram (ERD) illustrates the database structure of the Clinic Appointment and Patient Management System. It is composed of six tables: patients, doctors, specialties, appointments, consultations, and prescriptions. Each table is responsible for storing a particular set of information needed for managing patient records, doctor information, appointment schedules, medical consultations, and prescribed medications. The appointments table plays an important role because it connects patients with doctors and serves as the starting point for recording consultation details.

The patients table contains the personal and contact information of individuals who receive medical services from the clinic. Its attributes include patient_id (primary key), full_name, date_of_birth, gender, address, contact_number, email, and emergency_contact. The patient_id uniquely identifies every patient in the database. This table is connected to the appointments table through patient_id, establishing a one-to-many relationship. This means that one patient can schedule several appointments, while each appointment belongs to only one patient.

The doctors table stores the information of medical professionals working at the clinic. It includes doctor_id (primary key), specialty_id (foreign key), full_name, license_number,

email, contact_number, and consultation_fee. Each doctor is assigned to a particular medical specialty through specialty_id. The relationship between doctors and specialties is one-to-many because one specialty can be associated with several doctors, but each doctor belongs to only one specialty. This connection allows the system to organize doctors according to their respective medical fields.

The specialties table maintains the different areas of medical practice available in the clinic. It contains specialty_id (primary key) and specialty_name. The specialty_id serves as the unique identifier for every specialty. Through its relationship with the doctors table, the system can determine which medical field a doctor belongs to. This organization also makes it easier to identify doctors based on the services and expertise they provide.

The appointments table manages the scheduled visits between patients and doctors. It contains appointment_id (primary key), patient_id and doctor_id (foreign keys), appointment_date, appointment_time, reason_for_visit, appointment_status, and notes. The foreign keys connect each appointment to its corresponding patient and assigned doctor. Both patients and doctors have one-to-many relationships with appointments. A patient may have multiple scheduled visits, and a doctor may attend to multiple patients through separate appointments. The appointments table therefore acts as the connecting entity between patients and doctors. It also stores important scheduling information and indicates whether an appointment is scheduled, completed, cancelled, or marked as a no-show. The table also has a composite unique constraint (uq_doctor_schedule) on doctor_id, appointment_date, and appointment_time. This prevents a doctor from being booked for two appointments at the same date and time, which enforces the business rule that a doctor cannot have overlapping appointments.

The consultations table records the medical information gathered during a patient's consultation with a doctor. Its attributes include consultation_id (primary key), appointment_id (foreign key), symptoms, diagnosis, medical_notes, treatment_recommendations, and consultation_date. The appointment_id is also assigned a unique constraint, ensuring that an appointment can be associated with only one consultation record. This creates a one-to-one relationship between appointments and consultations. Through this connection, the clinic can refer to the appointment where a particular medical assessment took place and review the information recorded during the consultation.

The prescriptions table stores the medications and instructions provided as part of a patient's treatment. It consists of prescription_id (primary key), consultation_id (foreign key),

medication, dosage, frequency, duration, and instructions. Its relationship with the consultations table is one-to-many because a single consultation may result in several prescribed medications. However, each prescription is linked to only one consultation. This allows the system to keep medication records connected to the appropriate medical assessment.

Overall, the database structure organizes the clinic's information through connected tables, allowing patient records, doctor assignments, medical specialties, appointments, consultations, and prescriptions to be managed in an organized manner. The appointments table connects patients and doctors, while the consultations and prescriptions tables maintain the medical information associated with each visit. The use of primary keys, foreign keys, and relationship constraints helps maintain consistency between records and makes it easier to retrieve information when needed. This design provides a structured foundation for handling the clinic's appointment scheduling and patient management activities.

SQL Output

1. Display All Patients and Their Appointment History

```
SELECT

    P.patient_id,
    P.full_name AS patient_name,

    a.appointment_id,
    a.appointment_date,
    a.appointment_time,

    d.full_name AS doctor_name,
    a.reason_for_visit,
    a.appointment_status
FROM patients as p
LEFT JOIN appointments as a
    ON a.patient_id = p.patient_id
LEFT JOIN doctors as d
    ON a.doctor_id = d.doctor_id
ORDER BY p.patient_id, a.appointment_date;
```

patient_id ▲ 1	patient_name	appointment_id	appointment_date	appointment_time	doctor_name	reason_for_visit	appointment_status
1	Maria Santos	1	2026-11-04	09:00:00	Dr. Denzemiel Ethan S. Dacillo	Fever and cough for 3 days	Completed
1	Maria Santos	11	2026-11-06	11:30:00	Dr. Jelres C. Romano	Annual Acne treatment	Completed
2	Juan Dela Cruz	2	2026-11-10	10:30:00	Dr. Christoff Allen B. Jumapit	Chest discomfort	Scheduled
2	Juan Dela Cruz	12	2026-11-13	14:30:00	Dr. Elena Cruz	Headache and dizziness	Completed
3	Ana Reyes	3	2026-11-05	13:00:00	Dr. Denemiel Nathan S. Dacillo	Child vaccination	Completed
4	Jose Rizal Jr.	4	2026-11-07	09:30:00	Dr. John Carl T. Acibo	Knee pain after jogging	Completed
5	Liza Villanueva	5	2026-11-04	11:00:00	Dr. Jelres C. Romano	Skin rash and itching	Completed
6	Mark Bautista	6	2026-11-11	14:00:00	Dr. Denemiel Nathan S. Dacillo	Stomach ache and fever	Scheduled
7	Sofia Lim	7	2026-11-10	09:00:00	Dr. Ildefonso D. Providencia	Prenatal check-up	Scheduled
8	Carlos Mendoza	8	2026-11-12	10:00:00	Dr. Denzemiel Ethan S. Dacillo	MonthlyPhysical exam	Completed
9	Grace Tan	9	2026-11-12	15:00:00	Dr. Christoff Allen B. Jumapit	High blood pressure follow-up	Scheduled
10	Paolo Garcia	10	2026-11-10	09:00:00	Dr. John Carl T. Acibo	Basketball Ankle sprain	Scheduled

2. Display Doctors' Daily Schedule

```sql
SELECT
    d.doctor_id,
    d.full_name AS doctor_name,
    s.specialty_name,
    a.appointment_date,
    a.appointment_time,
    p.full_name AS patient_name,
    a.reason_for_visit,
    a.appointment_status
FROM doctors as d
INNER JOIN specialties as s
    ON d.specialty_id = s.specialty_id
INNER JOIN appointments as a
    ON d.doctor_id = a.doctor_id
INNER JOIN patients as p
    ON a.patient_id = p.patient_id
WHERE a.appointment_date = '2026-11-10'
ORDER BY d.full_name, a.appointment_time;
```

Extra options

doctor_id	doctor_name	specialty_name	appointment_date	appointment_time	patient_name	reason_for_visit	appointment_status
3	Dr. Christoff Allen B. Jumapit	Cardiology	2026-11-10	10:30:00	Juan Dela Cruz	Chest discomfort	Scheduled
5	Dr. Ildefonso D. Providencia	Obstetrics and Gynecology	2026-11-10	09:00:00	Sofia Lim	Prenatal check-up	Scheduled
6	Dr. John Carl T. Acibo	Orthopedics	2026-11-10	09:00:00	Paolo Garcia	Basketball Ankle sprain	Scheduled

Show all | Number of rows: 25 | Filter rows: Search this table | Sort by key: None

3. Display Patient Consultations, Diagnoses, and Prescriptions

Run SQL query/queries on database clinic_db:

```sql
SELECT p.full_name as patient_name,
    d.full_name as doctor_name,
    c.consultation_date, c.symptoms, c.diagnosis, c.treatment_recommendations,
    pr.medication, pr.dosage, pr.frequency, pr.duration, pr.instructions
FROM patients as p
INNER JOIN appointments as a
    ON p.patient_id = a.patient_id
INNER JOIN doctors as d
    ON a.doctor_id = d.doctor_id
INNER JOIN consultations as c
    ON a.appointment_id = c.appointment_id
LEFT JOIN prescriptions as pr
    ON c.consultation_id = pr.consultation_id
ORDER BY c.consultation_date, p.full_name;
```

patient_name	doctor_name	consultation_date ▲ 1	symptoms	diagnosis	treatment_recommendations
Liza Villanueva	Dr. Jelres C. Romano	2026-11-04	Red itchy patches on arms	Allergic contact dermatitis	Avoid trigger, antihistamine, topical cream
Liza Villanueva	Dr. Jelres C. Romano	2026-11-04	Red itchy patches on arms	Allergic contact dermatitis	Avoid trigger, antihistamine, topical cream
Maria Santos	Dr. Denzemiel Ethan S. Dacillo	2026-11-04	Fever, cough, sore throat	Viral upper respiratory infection	Rest, fluids, paracetamol as needed
Juan Dela Cruz	Dr. Christoff Allen B. Jumapit	2026-11-10	Chest tightness, shortness of breath on exertion	Stable angina, Hypertension	Low-salt diet, continue maintenance meds, follow-u...
Juan Dela Cruz	Dr. Christoff Allen B. Jumapit	2026-11-10	Chest tightness, shortness of breath on exertion	Stable angina, Hypertension	Low-salt diet, continue maintenance meds, follow-u...
Carlos Mendoza	Dr. Denzemiel Ethan S. Dacillo	2026-11-12	No symptoms, for routine check	Normal physical exam	Continue healthy lifestyle, annual labs
Grace Tan	Dr. Christoff Allen B. Jumapit	2026-11-12	Headache, occasional dizziness	Uncontrolled hypertension	Adjust medication, low-salt diet, BP monitoring
Juan Dela Cruz	Dr. Elena Cruz	2026-11-13	Headache, dizziness, fatigue	Dehydration and migraine	Increase fluid intake, rest, pain reliever as need...

medication	dosage	frequency	duration	instructions
Cetirizine	10mg	Once daily at bedtime	7 days	May cause drowsiness
Hydrocortisone Cream 1%	Apply thin layer	Twice daily	7 days	Apply only on affected area, do not use on face
Paracetamol	500mg	Every 6 hours as needed	3 days	Take after meals if fever or pain persists
Aspirin	80mg	Once daily	30 days	Take after meals
Amlodipine	5mg	Once daily	30 days	Take at the same time daily, monitor BP
Multivitamins	1 tablet	Once daily	30 days	Take after breakfast
Losartan	50mg	Once daily	30 days	Take in the morning, check BP weekly
Ibuprofen	400mg	Every 8 hours as needed	3 days	Take with food, stop if stomach pain occurs